problem_2a

- Data $\quad I \in \{1,2,3\}$

  C_i^c = cost per computer from Vendor $i \in I$

  C_i^d = cost of delivery from Vendor $i \in I$.

  Q_i^{max} = the most quantity of computers to be sold from Vendor $i \in I$

  Q^{min} = the least quantity of computers to be ordered from Vendor $i \in I$

  D = the total demand of computers of State University

- Decision

  x_i = the quantity of computers ordered from Vendor $i \in I$

  $y_i = \begin{cases} 1, & \text{if State University orders computer from Vendor } i \in I. \\ 0, & \text{otherwise} \end{cases}$

- Model

$$\min \sum_{i=1}^{3} C_i^c x_i + \sum_{i=1}^{3} C_i^d y_i \qquad \text{minimize the cost of computers and delivery}$$

$$\text{s.t.} \quad \sum_{i=1}^{3} x_i = D \qquad \text{the total demand}$$

$$x_i \le Q_i^{max} y_i \quad \forall i \in I \qquad \text{the most quantity}$$

$$x_i \ge Q^{min} y_i \quad \forall i \in I \qquad \text{the least quantity}$$

$$x_i \ge 0 \text{ and Integer} \quad \forall i \in I$$

$$y_i \text{ binary} \quad \forall i \in I$$